COURSE 2 · SECTION 4 · EXERCISE

Late drawings, findings, mass and cost

Section 4 · Drawing release, review findings, mass/cost vs target

Start: Section05_1_Start.pbix or your Engxtech_S04.pbix; save a copy as Engxtech_S04_Exercise.pbix

Time: about 20 min

Data: the course data folder

DESIGN QUESTION

How often are drawings released on time, what do design reviewers find most often, and which part revisions miss their mass and cost targets?

Extra practice. This exercise is extra practice: a longer task than the video's short "Your turn" questions, which are answered in the video itself. Do it after the video, in a copy of your file.

Drawings has the planned and actual drawing release (sign-off) dates per drawing sheet, not the revision effective date in Revisions; the video added DaysLate and the release measures. ReviewFindings links to Drawings by DrawingNo, Drawings to Revisions by PartRev. MassCost holds the actual mass and cost per part revision; the targets are in Parts.

New in this exercise (not shown in the video; one line each):

- **ALL (column)**: inside CALCULATE, removes the filter on that column, so the total ignores the Category on the axis.

Your tasks

1. In your exercise copy, read the Drawings page: **On-time %**, **Avg Days Late** and the P300-B row (DaysLate, ReviewRounds).
2. Findings page: read the Pareto (Finding Count columns, **Findings Cum %** line) and the Finding Count and Major Findings cards.
3. New measure in _Measures: **Tolerance Share** = Tolerance findings / all findings, so it ignores the Category on the axis.
4. Targets page: read Revs Over Mass Target, Revs Over Cost Target and the largest mass and cost overruns.
5. Check that the two bar charts are labelled clearly: one for mass (%), one for cost (%). Do not mix them on one axis.

Check your answer

Check	Expected
On-time release	**33.3 % (7/21)**
Average lateness of the late drawings	**4.6 d**
P300-B drawing: days late / review rounds	**19 d / 3**
Findings / of which major	**60 / 19**
Top Pareto bar	**Tolerance 23**
Tolerance Share	**38.3 %**
Cumulative % after the first three bars	**71.7 %**
Revisions over mass target	**15 of 16**
Revisions over cost target	**10 of 16**
Largest mass / cost overrun	**P100-B +12.4 % / P400-A +21.8 %**

Hint. GD&T and Material tie at 10 findings each. Findings Cum % ranks by count, so both bars get the same cumulative value, which is correct: check the value after the third bar.

Stuck? The worked solution with the full DAX and M is in Solutions/S04_Solution.pdf.